



U.S. Department
of Transportation

**Research and
Special Programs
Administration**

JUN 2 1998

400 Seventh Street, S.W.
Washington, D.C. 20590

DOT-E 11266
(SECOND REVISION)

EXPIRATION DATE: May 31, 2000

1. GRANTEE: AEA Technology/QSA, Inc. (formerly Amersham Corporation) Burlington, Massachusetts, and other persons offering radioactive material packages under specific instructions provided to them by AEA Technology/QSA, Inc.
2. PURPOSE AND LIMITATION: This exemption authorizes transport in commerce of devices containing Class 7 (radioactive material) sources in a DOT Specification 20WC (20WC) overpack without valid documentation that the devices meet Type A or Type B radioactive material package standards, or that the encapsulated sources meet special form standards; and provides no relief from any regulation other than those specifically stated.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR 173.416(e) and 173.476.
5. BASIS: This exemption is based on AEA Technology/QSA, Inc.'s application dated April 2, 1998 in accordance with 49 CFR 107.109.
6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous materials description/proper shipping name	Hazard Class number	Identification number	Packing Group
Radioactive material, special form, n.o.s.	Class 7	UN2974	N/A

7. SAFETY CONTROL MEASURES:a. PACKAGING -

- (1) The Class 7 (radioactive material) sources that may be transported under this exemption are encapsulations containing Co-60, Cs-137, or Am-241- beryllium neutron sources that were classified as special form in the past. These sources must be contained in devices such as industrial radiography or other similar controlled irradiation devices that were formerly approved or licensed by the Nuclear Regulatory Commission or by State agencies with licensing authority. At present, the individual sources may be either a Type A or Type B quantity, but not more than 50 Ci of Co-60 and/or Cs-137, or 20 Ci of Am-241 may be transported in any one package.
- (2) The devices containing the sources may be transported in 20WC packages that meet all of the provisions of 49 CFR 178.362. If the 20WC is not provided by AEA Technology/QSA, Inc., the shipper shall provide AEA Technology/QSA, Inc. with a description of the 20WC to be used. The 20WC package shall be prepared for transport following the written instructions prepared by AEA Technology/QSA, Inc. Special attention shall be given to ensuring that the device mechanisms are secured so that the radiation levels outside the devices are low, and packaging/cushioning materials within the cavity of the 20WC shall prevent movement of the devices and source controls during transport.

b. TESTING -

- (1) The encapsulated sources do not need to have the evidence of satisfying special form conditions as normally required by 49 CFR 173.476 for allowing the shipments to be offered as Radioactive material, special form, n.o.s., UN2974. The devices do not need to be documented as meeting DOT Specification 55 or Specification 7A requirements as normally required by 49 CFR 173.416(e) or 173.415(a).
- (2) Within six months prior to any source or device being offered for transport in a 20WC, the shipper shall have evidence that there is no apparent

leakage of radioactive material from the source, and at the time of transport the outside surfaces of the device shall not have contamination levels exceeding the limits shown in the table of 49 CFR 173.443(a).

- c. MARKING - The 20WC shall be marked in accordance with Subpart D of Part 172.

8. SPECIAL PROVISIONS.

a. Persons possessing encapsulated sources and devices described in paragraph 7(a) may transport them to any authorized/licensed radioactive waste facility. Each shipment will be approved on a case by case basis, with AEA Technology/QSA, Inc. assuring that the waste facility recipient is authorized/qualified to receive the package. AEA Technology/QSA, Inc. will maintain control of the 20 Wcs and must give approval for each shipment that goes to a different location. Even though a shipper follows all of the instructions provided by AEA Technology/QSA, Inc., it is the shipper's responsibility to ensure that the shipment is in conformance with the radioactive material provisions of Parts 100 - 185 of 49 CFR, except for the relief provided by this exemption.

b. When the package leaves the shipper's facility, AEA Technology/QSA Inc. shall be notified of the particulars of the shipment and when it is expected to arrive at a radioactive waste facility.

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle.

10. MODAL REQUIREMENTS:

a. A copy of this exemption must be carried aboard each motor vehicle used to transport packages covered by this exemption.

11. COMPLIANCE. Failure by a person to comply with any of the following may result in suspension or revocation of this exemption and penalties prescribed by the hazardous materials transportation law, 49 U.S.C. Section 5101 et seq.:

- o All terms and conditions prescribed in this exemption and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Registration required by 49 CFR 107.106 et seq., when applicable.

JUN 2 1998

continuation of 2nd rev. DOT-E 11266

Page 4

Each "hazmat employee", as defined in 49 CFR 171.8, who performs a function subject to this exemption must receive training on the requirements and conditions of this exemption in addition to the training required by 49 CFR 172.700 through 172.704.

No person may use or apply this exemption, including display of its number, when the exemption has expired or is otherwise no longer in effect.

12. REPORTING REQUIREMENTS. The carrier is required to report any incident involving loss of packaging contents or packaging failure to the Associate Administrator for Hazardous Materials Safety (AAHMS) as soon as practicable. (49 CFR 171.15 and 171.16 apply to any activity undertaken under the authority of this exemption.) In addition, the holder(s) of this exemption must inform the AAHMS, in writing, of any incidents involving the package and shipments made under the terms of this exemption.

Issued at Washington, D.C.:

for Alan I. Roberts
Associate Administrator for
Hazardous Materials Safety

6/2/98
(DATE)

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Research and Special Programs Administration, Department of Transportation, Washington, D.C. 20590.
Attention: DHM-31.

The original of this exemption is on file at the above office. Photo reproductions and legible reductions of this exemption are permitted. Any alteration of this exemption is prohibited.

Dist: FHWA
PO: am